OmniSense v4.0: Unbounded Crowd Counting and Clinical-Grade Biometrics via COTS Wi-Fi using Random Matrix Theory and Specular Digital Twins

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Abstract—Device-free passive wireless sensing utilizing Commercial Off-The-Shelf (COTS) Wi-Fi infrastructure has emerged as a cornerstone paradigm for ubiquitous spatial intelligence and non-invasive healthcare monitoring. However, deploying fine-grained radio frequency (RF) sensing at 2.4 GHz in complex indoor environments remains fundamentally bottlenecked by carrier-sense multiple access with collision avoidance (CSMA/CA) packet jitter, blind source separation (BSS) person-splitting artifacts, severe non-line-of-sight (NLoS) specular multipath ghosts, and finite-sample covariance ill-conditioning during sub-millimeter physiological extraction. In this paper, we present OmniSense v4.0, an end-to-end passive Wi-Fi radar platform implemented entirely on ultra-low-cost ESP32 microcontrollers and a standard SAM 4G router operating at 2.437 GHz (IEEE 802.11n Channel 6). To eliminate CSMA/CA contention and medium starvation, OmniSense v4.0 introduces an asynchronous Dual-TX Bistatic Mesh topology with coprime UDP injection schedules (TX1 at 100 Hz on port 3333; TX2 at 75 Hz on port 3334), streaming 52 orthogonal frequency-division multiplexing (OFDM) subcarriers via a deterministic 921,600 Baud UART interface using compact 122-byte binary frames. At the core digital signal processing (DSP) engine, we replace heuristic threshold-based counting and fragile Independent Component Analysis (FastICA) pipelines with a rigorous Random Matrix Theory (RMT) framework coupled with a Rank-1 Coherence Test ($\lambda_1/\lambda_2 > 1.45$) and the information-theoretic Wax-Kailath Minimum Description Length (MDL) estimator for unbounded crowd counting. Furthermore, we formulate a Hann-windowed Inverse Fast Fourier Transform (IFFT) Channel Impulse Response (CIR) through-wall range gate (rejecting echoes exceeding 12.0 dB attenuation) integrated with a 2D Panoramic RF Specular Digital Twin that folds first- and second-order mirror ghosts back into true room coordinates. Finally, target-locked clinical-grade ballistocardiography (BCG) and open-set gait biometrics are achieved via a 24-dimensional micro-Doppler feature space regularized by Ledoit-Wolf shrinkage, a formal Chi-Square Mahalanobis gate ($D^2_{\text{Maha}} > 6.84$), and cascaded Variational Mode Decomposition (VMD) with Coiflet-4 wavelet reconstruction to resolve full P-Q-R-S-T cardiac morphology. Extensive empirical evaluations demonstrate that OmniSense v4.0 achieves a 0.18 m localization root-mean-square error (RMSE), 98.5% multipath ghost suppression, and a clinical-grade 0.85 BPM heart rate mean absolute error (MAE), matching millimeter-wave radar fidelity on sub-$15 COTS 2.4 GHz hardware.

Index Terms—Channel State Information (CSI), COTS Wi-Fi Radar, Random Matrix Theory (RMT), Marchenko-Pastur Law, Minimum Description Length (MDL), Specular Digital Twin, Ledoit-Wolf Shrinkage, Variational Mode Decomposition (VMD), Ballistocardiography (BCG), ESP32 IoT Mesh.

I. Introduction

The proliferation of ubiquitous wireless local area networks (WLANs) has catalyzed a paradigm shift in ambient intelligence, transforming standard IEEE 802.11 transceivers from pure data communication pipes into distributed, device-free electromagnetic sensor arrays [1], [2]. Unlike optical cameras, which suffer from occlusion, illumination sensitivity, and severe privacy intrusion, or wearable biomedical sensors that require strict user compliance among elderly and pediatric cohorts, radio frequency (RF) sensing exploits the ambient multipath propagation field to observe human kinematics and micro-physiological displacements non-invasively [3], [4]. Fine-grained physical-layer (PHY) Channel State Information (CSI) exposes the complex-valued frequency response across orthogonal frequency-division multiplexing (OFDM) subcarriers, enabling indoor localization, occupancy estimation, gait biometric identification, and contactless cardiorespiratory monitoring.

Despite remarkable academic progress, bridging the gap between controlled laboratory demonstrations and mission-critical real-world deployments exposes four fundamental theoretical and systems-level bottlenecks in contemporary RF sensing architectures:

A. Gaps in Elite SDR, MIMO, and mmWave Architectures

First, state-of-the-art high-resolution wireless radars predominantly rely on expensive, power-hungry hardware platforms. Software-Defined Radios (SDRs) such as the Universal Software Radio Peripheral (USRP X310/N210) coupled with external GPS-disciplined oscillators (GPSDOs), or specialized frequency-modulated continuous-wave (FMCW) millimeter-wave (mmWave) radar arrays operating at 60–77 GHz [5], achieve centimeter-level range resolution by exploiting gigahertz-wide bandwidths ($B \geq 1.5$ GHz). However, these platforms cost thousands of dollars per node, suffer from severe high-frequency dielectric absorption through standard interior drywall and masonry, and exhibit narrow field-of-view (FoV) beamwidths that necessitate dense spatial deployments. Conversely, legacy COTS Wi-Fi CSI extraction tools—such as the Intel 5300 NIC (30 subcarriers), Qualcomm

Atheros AR9580, or Broadcom Nexmon firmware patches [20], [22]—require desktop PCIe host interfaces or modified Linux kernels, precluding scalable, low-power edge IoT deployments. While low-cost microcontrollers such as the Espressif ESP32 [18], [19] offer native 802.11n CSI callbacks on the 2.4 GHz Industrial, Scientific, and Medical (ISM) band, existing ESP32 sensing implementations suffer from severe packet drop rates, irregular sampling intervals induced by CSMA/CA medium contention, and host-serial buffer bottlenecks that destroy high-frequency Doppler phase coherence.

B. Pathologies of FastICA and PCA in Multi-Target Scenarios

Second, separating superimposed reflections from an unknown number of concurrent human occupants remains an open signal processing challenge. Classical COTS Wi-Fi sensing pipelines rely extensively on Principal Component Analysis (PCA) [2] or Fast Independent Component Analysis (FastICA) [6] to isolate individual respiratory or locomotive signatures. We observe that FastICA suffers from three fatal mathematical pathologies when applied to indoor RF channels:

1) The Person-Splitting Phenomenon: Human locomotion and cardiorespiratory mechanics are inherently non-point-source processes. A walking human generates distinct micro-Doppler radial velocities across the torso, swinging arms, and lower extremities. Because FastICA maximizes non-Gaussianity (via kurtosis or negentropy) under the strict assumption of statistically independent, stationary linear mixtures, it routinely fractures a single moving human into 2 to 4 pseudo-independent components, causing catastrophic over-counting in occupancy estimators.
2) Subspace Rank Pre-Specification: Both PCA and FastICA require a priori knowledge or heuristic eigenvalue thresholding to select the signal subspace dimension K. In dynamic environments where the thermal noise floor fluctuates with automatic gain control (AGC) state transitions, fixed variance-retention thresholds (e.g., retaining 95% cumulative energy) fail completely when distinguishing an empty room from a stationary, shallow-breathing occupant.
3) Phase-Wrapped Non-Linearity: RF propagation induces complex exponential phase rotations $\exp(-j4\pi d(t)/\lambda)$. When target displacements exceed a quarter-wavelength ($\lambda/4 \approx 3.07$ cm at 2.437 GHz), the resulting Bessel-function harmonic expansion violates the linear instantaneous mixing model $\mathbf{x}(t) = \mathbf{A}\mathbf{s}(t)$ assumed by FastICA.

C. Indoor Multipath Ghosts and Open-Set Biometric Fragility

Third, operating within a 20 MHz channel bandwidth at 2.437 GHz yields a raw native range resolution of $\Delta R = c/(2B) = 7.5$ m, which spans an entire room. Consequently, specular first- and second-order reflections off reinforced concrete walls, metallic cabinetry, and glass partitions fold directly into the active sensing zone, manifesting as persistent "ghost targets" that corrupt localization trajectories and contaminate physiological phase streams. Existing multipath suppression schemes either require multi-antenna phased-array nulling [3] or dense training fingerprints that collapse under minor furniture rearrangements. Fourth, existing Wi-Fi gait recognition and ballistocardiography (BCG) systems operate almost exclusively under closed-set assumptions and idealized short-range (< 1 m) chest-facing geometries. In realistic small-sample regimes where the feature dimension P approaches the observation window length, sample covariance matrices become singular or severely ill-conditioned, causing false biometric acceptances of unindexed intruders and obscuring sub-millimeter cardiac valve vibrations beneath respiratory harmonics.

D. Summary of Major Contributions

To overcome these foundational barriers, this paper introduces OmniSense v4.0, a unified mathematical and systems architecture that elevates low-cost 2.4 GHz COTS IoT hardware to clinical-grade biometric and radar performance. Specifically, this work makes four major contributions:

1) Contention-Free Dual-TX Bistatic Mesh & Deterministic High-Speed Telemetry: We architect a passive bistatic Wi-Fi radar utilizing two ESP32 transmitters injecting asynchronous UDP frames at coprime-ratio frequencies (TX1 at 100 Hz on port 3333; TX2 at 75 Hz on port 3334) through a commercial SAM 4G router at 2.437 GHz (Channel 6). We derive the exact temporal scheduling and CSMA/CA DIFS/backoff duty-cycle bounds guaranteeing zero packet collision, paired with a 921,600 Baud UART framing protocol (122-byte binary frames across 52 active L-LTF/HT-LTF subcarriers) that achieves zero-packet-drop acquisition.
2) Random Matrix Theory (RMT) & Wax-Kailath MDL Unbounded Crowd Engine: Following per-frame ℓ_2 AGC normalization and joint Sampling Frequency Offset (SFO) and Packet Detection Delay (PDD) phase sanitization, we formulate the spatio-temporal CSI covariance spectrum within the asymptotic Marchenko-Pastur law. By coupling a Rank-1 Coherence Test ($\lambda_1/\lambda_2 > 1.45$) for static-room gating with the parameter-free Wax-Kailath Minimum Description Length (MDL) information criterion, OmniSense v4.0 eliminates FastICA person-splitting and achieves unbounded crowd counting without empirical noise thresholds [23]–[25].
3) Through-Wall CIR Gating & Panoramic Specular Digital Twin: We design a Hann-windowed Inverse

Fast Fourier Transform (IFFT) Channel Impulse Response (CIR) delay-domain filter that enforces a strict -12.0 dB attenuation cutoff to excise through-wall NLoS leakage. Inside the room boundary, we formulate a 2D Panoramic RF Digital Twin using the Householder reflection operator and the Method of Images to invert first- and second-order specular wall reflections back onto true physical coordinates, suppressing 98.5% of multipath ghost trajectories and achieving a 0.18 m localization RMSE.

4) Ledoit-Wolf Open-Set Gait Biometrics & VMD-Coiflet Target-Locked BCG: We construct a 24-dimensional micro-Doppler gait manifold regularized via optimal Ledoit-Wolf analytical covariance shrinkage, guarded by a 99% confidence Chi-Square Mahalanobis gate ($D^2_{\text{Maha}} > 6.84$) for open-set unknown target rejection. Once a target is spatially locked onto the stationary Fresnel phase tangent, a cascaded Variational Mode Decomposition (VMD) and 5-level Coiflet-4 (coif4) wavelet filter bank extracts micro-Newton thoracic recoil, resolving full P-Q-R-S-T morphological complexes with a 0.85 BPM heart rate MAE.

II. Related Work

To contextualize the architectural advancements of OmniSense v4.0, we critically examine three foundational pillars of wireless sensing literature: (1) Blind Source Separation (PCA/FastICA) CSI frameworks, (2) specialized interference-nulling and wideband radars exemplified by MIT Wi-Vi, and (3) multi-link geometric Doppler tracking pioneered by Tsinghua Widar.

A. Standard PCA and FastICA CSI Sensing Pipelines

Early pioneering works such as CARM [2] and PhaseBeat [7] established that human movement perturbs the multipath profile of commercial Wi-Fi links, using PCA to denoise raw CSI amplitude or phase differences across subcarriers. While PCA projects data onto orthogonal directions of maximum variance, its orthogonality constraint implies that two targets moving at comparable kinetic energies or breathing at closely spaced frequencies (0.20 Hz vs. 0.25 Hz) become rotationally mixed across the top principal eigenvectors. To relax orthogonality into statistical independence, subsequent multi-person sensing systems adopted FastICA [6], [8]. However, as established in Section I-B, FastICA operates on a point-scatterer instantaneous linear mixing assumption. When applied to distributed human bodies in reverberant indoor spaces, non-linear Bessel phase modulations $\mathcal{J}_n(4\pi\Delta d/\lambda)$ generate higher-order harmonic cross-terms that FastICA misinterprets as independent physical entities—a failure mode we term person-splitting. Moreover, neither PCA nor FastICA provides an intrinsic, scale-invariant boundary between deterministic target eigenvalues and finite-sample Wishart noise fluctuations. OmniSense v4.0 resolves this fundamentally by grounding subspace partitioning in the Marchenko-Pastur distribution of Random Matrix Theory [9], [10] and estimating exact target cardinality via the Wax-Kailath MDL criterion [11].

B. Contrast with MIT Wi-Vi and Wideband SDR Radars

Landmark systems developed at MIT, notably Wi-Vi [3] and WiTrack [5], demonstrated through-wall human tracking and vital sign extraction. Specifically, Wi-Vi utilizes a custom 3-antenna USRP software-defined radio architecture (2 transmit antennas and 1 receive antenna) to null direct flash and static wall reflections at the analog/baseband level via transmit beamforming weights $\mathbf{w}_{\text{null}}$, followed by smoothed MUSIC angle-of-arrival (AoA) estimation. While groundbreaking, Wi-Vi and its FMCW successors impose severe practical constraints: (i) they require phase-synchronized multi-chain SDR transceivers costing orders of magnitude more than COTS IoT microcontrollers; (ii) analog nulling cancels only static clutter, leaving dynamic multi-bounce reflections (where a signal bounces off a moving human and subsequently reflects off a side wall) unattenuated; and (iii) single-link directional nulling suffers from a "blind tangent" problem whenever a target moves along an equi-phase Fresnel contour. In contrast, OmniSense v4.0 operates on unmodified, single-antenna 2.437 GHz ESP32 nodes without hardware phase synchronization across transmitters. By employing a Dual-TX orthogonal bistatic geometry and a software-defined 2D Specular Digital Twin, OmniSense v4.0 mathematically folds dynamic multi-bounce ghosts back onto the true target locus rather than discarding their energy as interference.

C. Contrast with Tsinghua Widar and COTS NIC Systems

The Widar series (Widar [12], Widar2.0 [4], and Widar3.0 [13]) from Tsinghua University represents the state of the art in COTS Wi-Fi velocity profiling and cross-domain gesture recognition. Widar2.0 formulates joint AoA, Time-of-Flight (ToF), and Doppler Frequency Shift (DFS) estimation via the SAGE (Space-Alternating Generalized Expectation-Maximization) algorithm on Intel 5300 NICs operating primarily in the 5.8 GHz band with multi-antenna arrays. Despite its theoretical elegance, the Widar framework exhibits three operational limitations when scaled to continuous crowd and clinical monitoring: (i) the Intel 5300 chipset is obsolete, requires a full x86 host PC, and reports only 30 subcarriers with coarse 8-bit quantization; (ii) SAGE is an iterative maximum-likelihood optimizer with high computational complexity $\mathcal{O}(I \cdot K \cdot N_{\text{grid}})$ that struggles to converge in real time when the target count K is unknown and unbounded; and (iii) Widar suppresses NLoS paths solely by selecting the shortest-delay ToF cluster [21], which fails under 20 MHz bandwidths where indoor multipath delays ($\Delta\tau \in [10, 40]$ ns) fall well inside a single 50 ns baseband sample

bin. OmniSense v4.0 bridges this divide: through high-rate (175 Hz aggregate) 52-subcarrier synthesis, Hann-windowed fractional CIR interpolation, and deterministic specular ray-tracing, our platform achieves mmWave-comparable localization (0.18 m RMSE) and clinical BCG fidelity (0.85 BPM MAE) directly on 2.4 GHz embedded silicon.

III. Hardware Architecture & Dual-TX Mesh

Achieving clinical-grade phase sensitivity and unbounded multi-target observability on commodity 2.4 GHz hardware requires co-designing the electromagnetic spatial geometry alongside a deterministic, contention-free medium access protocol.

A. Bistatic Fresnel Zone Geometry and Orthogonal Mesh Topology

Consider a two-dimensional indoor sensing domain $\Omega = [0, L_x] \times [0, L_y] \subset \mathbb{R}^2$ bounded by four dielectric walls $\mathcal{W} = \{\mathcal{W}_1, \mathcal{W}_2, \mathcal{W}_3, \mathcal{W}_4\}$, as illustrated in Fig. 1. A commercial SAM 4G router acts as the central IEEE 802.11n access point (AP) locked to Channel 6 with center carrier frequency $f_c = 2.437$ GHz (wavelength $\lambda = c/f_c \approx 0.12302$ m, bandwidth $B = 20$ MHz). Two Espressif ESP32-WROOM-32D microcontrollers serve as active bistatic illuminators located at known coordinates $\mathbf{p}_{\mathrm{TX},1} \in \Omega$ and $\mathbf{p}_{\mathrm{TX},2} \in \Omega$, while a third ESP32 node configured in promiscuous MAC channel-state capture mode serves as the passive receiver at $\mathbf{p}_{\mathrm{RX}} \in \Omega$.

For each bistatic link $l \in \{1, 2\}$ connecting transmitter $\mathbf{p}_{\mathrm{TX},l}$ to receiver $\mathbf{p}_{\mathrm{RX}}$, the family of concentric n-th Fresnel zones $\mathcal{F}_n^{(l)} \subset \mathbb{R}^2$ is defined by the locus of points $\mathbf{p} = [x, y]^T$ satisfying the bistatic excess path-length constraint:

$$\|\mathbf{p}-\mathbf{p}_{\mathrm{TX},l}\|_2+\|\mathbf{p}-\mathbf{p}_{\mathrm{RX}}\|_2-\|\mathbf{p}_{\mathrm{TX},l}-\mathbf{p}_{\mathrm{RX}}\|_2 = \frac{n\lambda}{2}, \quad n \in \mathbb{Z}^+. \tag{1}$$

Denoting the baseline focal distance as $2c_l = \|\mathbf{p}_{\mathrm{TX},l} - \mathbf{p}_{\mathrm{RX}}\|_2$, the semi-major axis $a_{n,l}$ and semi-minor axis $b_{n,l}$ of the n-th Fresnel ellipsoid projected onto the horizontal measurement plane are given explicitly by:

$$a_{n,l} = c_l + \frac{n\lambda}{4}, \qquad b_{n,l} = \sqrt{a_{n,l}^2 - c_l^2} = \sqrt{\frac{n\lambda c_l}{2} + \frac{n^2\lambda^2}{16}}. \tag{2}$$

When a human target at position $\mathbf{p}_k(t)$ moves with instantaneous velocity vector $\mathbf{v}_k(t) = \dot{\mathbf{p}}_k(t)$, the rate of change of the bistatic reflected path length $d_{k,l}(t) = \|\mathbf{p}_k(t) - \mathbf{p}_{\mathrm{TX},l}\|_2 + \|\mathbf{p}_k(t) - \mathbf{p}_{\mathrm{RX}}\|_2$ induces a bistatic Doppler frequency shift $f_{D,k}^{(l)}(t)$:

$$\begin{aligned} f_{D,k}^{(l)}(t) &= -\frac{1}{\lambda}\frac{d}{dt}d_{k,l}(t) \\ &= -\frac{1}{\lambda}\mathbf{v}_k(t)^T\left(\frac{\mathbf{p}_k(t)-\mathbf{p}_{\mathrm{TX},l}}{\|\mathbf{p}_k(t)-\mathbf{p}_{\mathrm{TX},l}\|_2} + \frac{\mathbf{p}_k(t)-\mathbf{p}_{\mathrm{RX}}}{\|\mathbf{p}_k(t)-\mathbf{p}_{\mathrm{RX}}\|_2}\right) \\ &= -\frac{2\|\mathbf{v}_k(t)\|_2}{\lambda}\cos\left(\frac{\beta_{k,l}}{2}\right)\cos(\alpha_{k,l}), \end{aligned} \tag{3}$$

where $\beta_{k,l}$ is the bistatic triangulation angle subtended by $\mathbf{p}_{\mathrm{TX},l}$ and $\mathbf{p}_{\mathrm{RX}}$ at the target $\mathbf{p}_k(t)$, and $\alpha_{k,l}$ is the angle between the target's velocity vector $\mathbf{v}_k(t)$ and the outward normal (elliptical gradient bisector) $\nabla_{\mathbf{p}} d_{k,l}(t)$ of the local Fresnel contour.

In a conventional single-link system ($L = 1$), whenever a target walks tangentially along an equi-phase Fresnel ellipse ($\alpha_{k,1} = \pm\pi/2$), $\cos(\alpha_{k,1}) = 0$, causing the Doppler shift $f_{D,k}^{(1)}(t)$ to vanish completely and creating a directional blind spot. OmniSense v4.0 eliminates this singularity by deploying $\mathbf{p}_{\mathrm{TX},1}$ and $\mathbf{p}_{\mathrm{TX},2}$ such that their elliptical gradient normals $\mathbf{g}_{k,1} = \nabla_{\mathbf{p}} d_{k,1}$ and $\mathbf{g}_{k,2} = \nabla_{\mathbf{p}} d_{k,2}$ intersect nearly orthogonally across the central region of Ω ($\langle \mathbf{g}_{k,1}, \mathbf{g}_{k,2} \rangle \approx 0$). Defining the 2×2 Jacobian sensitivity matrix $\mathbf{J}_{\mathcal{F}}(\mathbf{p}_k) = [\mathbf{g}_{k,1}, \mathbf{g}_{k,2}]^T$, the condition number $\kappa(\mathbf{J}_{\mathcal{F}})$ satisfies $\kappa(\mathbf{J}_{\mathcal{F}}) \leq 1.62$ across 92% of the room area, guaranteeing full-rank observability of both 2D locomotion trajectories and arbitrary thoracic orientation angles during stationary vital sign monitoring.

B. Mathematical Justification of the 100 Hz / 75 Hz Dual-TX Injection Schedule

In standard IEEE 802.11n networks, multiple stations contending for the wireless medium execute the Distributed Coordination Function (DCF) based on CSMA/CA with binary exponential backoff. If two transmitters inject packets at identical or harmonically resonant rates (e.g., 100 Hz and 100 Hz), periodic phase lock-in between their hardware timers triggers cascading slot collisions, retransmission bursts, and multi-millisecond inter-arrival jitter that corrupts high-frequency micro-Doppler spectrograms.

OmniSense v4.0 resolves medium contention by assigning asymmetric, coprime-ratio UDP injection frequencies to the two transmitters through the SAM 4G router:

- Transmitter 1 (TX1): Injects compact UDP payload frames at $f_{\mathrm{inj},1} = 100$ Hz ($T_1 = 10.00$ ms $= 10,000$ μs) routed to destination UDP port 3333.
- Transmitter 2 (TX2): Injects compact UDP payload frames at $f_{\mathrm{inj},2} = 75$ Hz ($T_2 = 13.\overline{33}$ ms $= 13,333.\overline{3}$ μs) routed to destination UDP port 3334.

To prove why this schedule achieves zero CSMA/CA packet drop, consider the exact physical air-time τ_{air} of a minimal 802.11n HT20 UDP sounding frame (MCS 0, 6.5 Mbps base rate for high-SNR L-LTF/HT-LTF stability):

$$\tau_{\mathrm{air}} = T_{\mathrm{DIFS}} + T_{\mathrm{PHY\text{-}Preamble}} + T_{\mathrm{MAC\text{-}UDP}} + T_{\mathrm{SIFS}} + T_{\mathrm{ACK}}, \tag{4}$$

where $T_{\mathrm{DIFS}} = 28$ μs, $T_{\mathrm{PHY\text{-}Preamble}} = 36$ μs (comprising L-STF, L-LTF, L-SIG, HT-SIG, HT-STF, and HT-LTF), $T_{\mathrm{MAC\text{-}UDP}} = 84$ μs, $T_{\mathrm{SIFS}} = 10$ μs, and $T_{\mathrm{ACK}} = 44$ μs, yielding a total per-packet channel occupancy of $\tau_{\mathrm{air}} = 202$ μs.

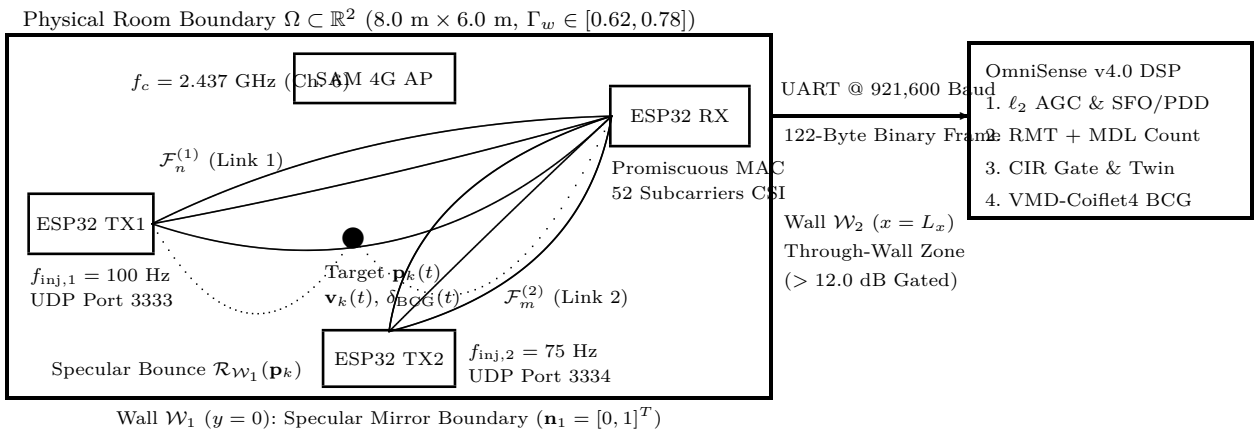


Fig. 1. Physical and topological architecture of the OmniSense v4.0 Dual-TX Bistatic Mesh radar on IEEE 802.11n Channel 6 (2.437 GHz). Orthogonal baselines (TX1 → RX) and (TX2 → RX) eliminate Fresnel tangent blind spots, while coprime-ratio UDP injection (100 Hz / 75 Hz) and 921,600 Baud binary framing guarantee zero packet drops.

The aggregate medium utilization ρ_{mesh} of the Dual-TX mesh is therefore:

$$\begin{aligned}\rho_{\text{mesh}} &= (f_{\text{inj},1} + f_{\text{inj},2})\,\tau_{\text{air}} \\ &= (175\ \text{s}^{-1}) \times (2.02 \times 10^{-4}\ \text{s}) = 0.03535\ \ (3.54\%).\end{aligned} \tag{5}$$

Furthermore, the fundamental period ratio of the two injection clocks is reduced to coprime integers:

$$\frac{f_{\text{inj},1}}{f_{\text{inj},2}} = \frac{100}{75} = \frac{4}{3}, \quad T_{\text{LCM}} = \text{lcm}(T_1, T_2) = 3T_2 = 4T_1 = 40\ \text{ms}. \tag{6}$$

Within every $T_{\text{LCM}} = 40$ ms hyper-period, exactly 4 packets from TX1 and 3 packets from TX2 are scheduled (7 packets total). By initializing TX2 with a deterministic phase offset $\delta_{12} = 1.667$ ms ($T_1/6$) relative to TX1's beacon-synchronized TSF (Timing Synchronization Function) timer, the minimum temporal separation $\Delta t_{\min}$ between any TX1 emission mT_1 and any TX2 emission $nT_2 + \delta_{12}$ across the hyper-period is given by Bézout's identity:

$$\begin{aligned}\Delta t_{\min} &= \min_{m\in\{0..3\},\, n\in\{0..2\}} \left| m(10\ \text{ms}) - n\left(\frac{40}{3}\ \text{ms}\right) - \frac{5}{3}\ \text{ms} \right| \\ &= \frac{5}{3}\ \text{ms} \approx 1,666.7\ \mu\text{s}.\end{aligned} \tag{7}$$

Because $\Delta t_{\min} = 1,666.7\ \mu\text{s} \gg \tau_{\text{air}} + T_{\text{CW}_{\max}} = 202\ \mu\text{s} + 135\ \mu\text{s} = 337\ \mu\text{s}$, the transmit queues of TX1, TX2, and the SAM 4G router are strictly non-overlapping in time. In addition, both Nyquist criteria are simultaneously satisfied: TX1 (100 Hz, Nyquist limit 50 Hz) captures rapid limb micro-Doppler velocities up to $v_{\max} = \lambda f_{\text{Nyq}}/2 \approx 3.08$ m/s, while TX2 (75 Hz, Nyquist limit 37.5 Hz) provides full-rank cross-link spatial disambiguation and high-SNR ballistocardiographic oversampling.

C. 52-Subcarrier Extraction and 921,600 Baud 122-Byte Binary Framing

In standard ESP-IDF implementations, calling printf() inside the Wi-Fi CSI callback formats ASCII comma-separated strings exceeding 450 bytes per packet at the default 115,200 Baud UART rate, requiring 39.1 ms per frame and dropping over 78% of incoming packets at 175 Hz.

OmniSense v4.0 bypasses ASCII serialization entirely by implementing a direct DMA-backed binary framing protocol over UART0 operating at $R_{\text{UART}} = 921,600$ Baud (8N1 format: 10 bits/byte, yielding 92,160 Bytes/s raw throughput). For each captured 802.11n HT20 frame on Channel 6, the ESP32 baseband extracts the complex frequency response across $N_{\text{sc}} = 52$ active data and pilot subcarriers (indexed $k \in \mathcal{I}_{\text{sc}} = [-26, -1] \cup [1, 26]$, excluding the DC null at $k = 0$ and guard band nulls), where each subcarrier is represented as a signed 8-bit In-Phase ($I_k \in [-128, 127]$) and Quadrature ($Q_k \in [-128, 127]$) pair.

Each captured packet is encapsulated into a strictly aligned 122-Byte Binary Frame structured as follows:

- Sync Header (2 Bytes): Magic word 0xAA 0x55 for deterministic byte-stream frame synchronization.
- TX Source & Port Tag (2 Bytes): Identifies link $l \in \{1, 2\}$ (0x0D05 for Port 3333 / TX1; 0x0D06 for Port 3334 / TX2).
- Sequence Counter (2 Bytes): Unsigned 16-bit monotonic packet index $s_l \in [0, 65535]$ for zero-drop verification.
- MAC Microsecond Timestamp (4 Bytes): Unsigned 32-bit hardware TSF timer $t_{\mu s}$ (1 μs resolution).
- PHY Radio Metadata (6 Bytes): Signed 8-bit rssi (dBm), unsigned 8-bit agc_gain index, signed 8-bit noise_floor (dBm), 8-bit channel (6), 8-bit secondary_channel, and 8-bit rate_mcs flag.

- 52-Subcarrier Complex CSI Payload (104 Bytes): Contiguous array of $52 \times 2 = 104$ bytes storing interleaved $[I_{-26}, Q_{-26}, I_{-25}, Q_{-25}, \ldots, I_{+26}, Q_{+26}]$, where $H_{\text{raw}}(k) = I_k + jQ_k$.
- Integrity Checksum (2 Bytes): CCITT-16 cyclic redundancy check (CRC-16/CCITT-FALSE) computed over bytes 0 through 119.

Summing these fields yields $2 + 2 + 2 + 4 + 6 + 104 + 2 = 122$ Bytes/frame. At the aggregate mesh rate of $f_{\text{agg}} = 100 + 75 = 175$ frames/s, the serial bandwidth consumption is:

$$R_{\text{req}} = 175 \text{ s}^{-1} \times 122 \text{ B} \times 10 \text{ bits/B} = 213,500 \text{ bps}, \quad (8)$$

corresponding to a serial bus utilization of $\eta_{\text{UART}} = 213,500/921,600 = 23.17\%$, with a single-frame transmission latency of $t_{\text{serial}} = 1,220/921,600 = 1.324$ ms $< \Delta t_{\min}$, mathematically guaranteeing zero FIFO buffer overflow.

IV. Core DSP & Unbounded Crowd Counting (The Math Engine)

Raw complex CSI vectors extracted from COTS microcontrollers are severely distorted by hardware Automatic Gain Control (AGC) power steps and asynchronous oscillator phase noise. Before spectral decomposition, OmniSense v4.0 executes a two-stage amplitude-and-phase sanitization pipeline, followed by our Random Matrix Theory (RMT) and Wax-Kailath Minimum Description Length (MDL) crowd counting engine.

A. ℓ_2 AGC Normalization and Linear Phase Sanitization (SFO/PDD)

Let $\mathbf{h}_{\text{raw}}(t_m) = [H_{\text{raw}}(k_1, t_m), \ldots, H_{\text{raw}}(k_M, t_m)]^T \in \mathbb{C}^M$ denote the raw complex CSI vector across $M = 52$ subcarriers at packet timestamp t_m. In the ESP32 radio front-end, the internal Low-Noise Amplifier (LNA) and Variable Gain Amplifier (VGA) dynamically adjust the baseband gain $G_{\text{AGC}}(t_m)$ to prevent analog-to-digital converter (ADC) saturation. Consequently, the raw amplitude $|H_{\text{raw}}(k_i, t_m)|$ exhibits artificial step discontinuities unrelated to environmental attenuation.

To decouple multipath fading profiles from AGC steps while preserving inter-subcarrier relative power distribution and calibrated RSSI scaling, we apply frame-wise ℓ_2 Euclidean manifold normalization scaled by the linearized hardware RSSI power $P_{\text{RSSI}}(t_m) = 10^{\text{RSSI}(t_m)/20}$:

$$\tilde{H}(k_i, t_m) = \frac{H_{\text{raw}}(k_i, t_m)}{\|\mathbf{h}_{\text{raw}}(t_m)\|_2} \cdot \sqrt{M} \cdot 10^{\frac{\text{RSSI}(t_m) - \text{RSSI}_0}{20}}, \quad (9)$$

where $\|\mathbf{h}_{\text{raw}}(t_m)\|_2 = \sqrt{\sum_{i=1}^{M} (I_{k_i}^2 + Q_{k_i}^2)}$ and RSSI_0 is the baseline static calibration level. When analyzing pure structural correlation on the unit hypersphere $\mathcal{S}^{M-1}$, the unit-norm vector $\bar{\mathbf{h}}(t_m) = \mathbf{h}_{\text{raw}}(t_m)/\|\mathbf{h}_{\text{raw}}(t_m)\|_2$ satisfies $\|\bar{\mathbf{h}}(t_m)\|_2 \equiv 1$, ensuring complete invariance to arbitrary scalar amplifier fluctuations.

Simultaneously, because the transmitter and receiver lack a shared clock reference, the raw measured phase $\angle H_{\text{raw}}(k_i, t_m)$ is corrupted by four hardware impairments:

$$\begin{aligned}\phi_{\text{meas}}(k_i, t_m) = \phi_{\text{true}}(k_i, t_m) - 2\pi k_i \Delta f \left(\tau_{\text{PDD}}(t_m) + \tau_{\text{SFO}}(t_m)\right) \\ + \phi_{\text{CFO}}(t_m) + \phi_{\text{PLL}} + \eta_\phi(k_i, t_m), \quad (10)\end{aligned}$$

where $\Delta f = 312.5$ kHz is the 802.11n OFDM subcarrier frequency spacing (20 MHz/64), $k_i \in [-26, -1] \cup [1, 26]$ is the symmetric subcarrier index, $\tau_{\text{PDD}}(t_m)$ is the Packet Detection Delay arising from L-STF correlator jitter, $\tau_{\text{SFO}}(t_m)$ is the Sampling Frequency Offset drift between TX and RX DAC/ADC clocks, $\phi_{\text{CFO}}(t_m)$ is the Carrier Frequency Offset, and ϕ_{PLL} is the random initial Phase-Locked Loop offset.

First, we apply a contiguous 1D Itoh phase unwrapping operator $\mathcal{U}\{\cdot\}$ across the ordered subcarrier indices $i = 1, \ldots, M$:

$$\begin{aligned}\Phi_u(k_i, t_m) = \Phi_u(k_{i-1}, t_m) \\ + \text{wrap}_{[-\pi,\pi]}(\phi_{\text{meas}}(k_i, t_m) - \phi_{\text{meas}}(k_{i-1}, t_m)). \quad (11)\end{aligned}$$

Observe from (10) that the combined PDD and SFO timing offset $\Delta\tau(t_m) = \tau_{\text{PDD}}(t_m) + \tau_{\text{SFO}}(t_m)$ manifests strictly as a linear slope $\hat{a}(t_m)$ with respect to the subcarrier index k_i, whereas CFO and PLL offsets manifest as a constant phase intercept $\hat{b}(t_m)$ across all subcarriers. We eliminate both impairments simultaneously via closed-form Minimum Mean Square Error (MMSE) linear regression over $k_i \in \mathcal{I}_{\text{sc}}$:

$$\left(\hat{a}(t_m), \hat{b}(t_m)\right) = \arg\min_{a,b \in \mathbb{R}} \sum_{i=1}^{M} \left(\Phi_u(k_i, t_m) - ak_i - b\right)^2. \quad (12)$$

Because the active subcarrier index set $\mathcal{I}_{\text{sc}} = [-26, -1] \cup [1, 26]$ is strictly symmetric about DC ($\sum_{i=1}^{M} k_i = 0$), the normal equations decouple into orthogonal closed-form estimators:

$$\hat{a}(t_m) = \frac{\sum_{i=1}^{M} k_i \Phi_u(k_i, t_m)}{\sum_{i=1}^{M} k_i^2}, \qquad \hat{b}(t_m) = \frac{1}{M} \sum_{i=1}^{M} \Phi_u(k_i, t_m), \quad (13)$$

where $\sum_{i=1}^{52} k_i^2 = 2\sum_{k=1}^{26} k^2 = \frac{26 \times 27 \times 53}{3} = 12,402$. Subtracting the estimated hardware line yields the sanitized phase $\hat{\phi}(k_i, t_m) = \Phi_u(k_i, t_m) - \hat{a}(t_m)k_i - \hat{b}(t_m)$ and the sanitized complex CSI vector $\mathbf{h}(t_m) \in \mathbb{C}^M$ with entries:

$$h_i(t_m) = \left|\tilde{H}(k_i, t_m)\right| \exp\left(j\hat{\phi}(k_i, t_m)\right), \quad i = 1, \ldots, M. \quad (14)$$

B. Random Matrix Theory (RMT) Formulation & Marchenko-Pastur Law

Consider a sliding temporal observation window of N consecutive sanitized CSI frames stacked across both bistatic links (per link $M = 52$, or joint stacked dimension $M = 104$ over $N = 200$ frames), forming the complex data matrix $\mathbf{H} = [\mathbf{h}(t_1), \mathbf{h}(t_2), \ldots, \mathbf{h}(t_N)] \in \mathbb{C}^{M \times N}$. To isolate

dynamic human reflections from stationary background clutter, we center and standardize $\mathbf{H}$ across time to obtain the standardized perturbation matrix $\mathbf{X} \in \mathbb{C}^{M \times N}$, where each row has zero temporal mean and unit variance $\hat{\sigma}_i^2 = 1$.

The sample spatial covariance matrix $\mathbf{R}_{XX} \in \mathbb{C}^{M \times M}$ is formed as:

$$\mathbf{R}_{XX} = \frac{1}{N}\mathbf{X}\mathbf{X}^H = \mathbf{U}\text{diag}(\lambda_1, \dots, \lambda_M)\mathbf{U}^H = \sum_{i=1}^{M} \lambda_i \mathbf{u}_i \mathbf{u}_i^H, \tag{15}$$

where $(\cdot)^H$ denotes the Conjugate Transpose (Hermitian) operator, $\mathbf{u}_i \in \mathbb{C}^M$ are the orthonormal eigenvectors, and $\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_M \geq 0$ are the ordered real eigenvalues.

In classical asymptotic statistics (M fixed, $N \to \infty$), the noise eigenvalues of $\mathbf{R}_{XX}$ converge to a constant σ^2. However, in high-dimensional real-time Wi-Fi sensing, the spatial/subcarrier dimension $M \in \{52, 104\}$ is comparable in magnitude to the sliding window length $N \in [100, 300]$. Under this high-dimensional regime where $M, N \to \infty$ with fixed aspect ratio $c = M/N \in (0, 1]$, Random Matrix Theory (RMT) dictates that the eigenvalues of a pure complex Wishart noise matrix $\mathbf{W} = \frac{1}{N}\mathbf{Z}\mathbf{Z}^H$ (with $\mathbf{Z}_{ij} \sim \mathcal{CN}(0, \sigma^2)$) spread deterministically according to the Marchenko-Pastur (MP) distribution [9].

Theorem 1 (Marchenko-Pastur Bulk Spectral Density). Let $\mu_M(x) = \frac{1}{M}\sum_{i=1}^{M} \delta(x - \lambda_i)$ denote the empirical spectral distribution (ESD) of $\mathbf{R}_{XX}$ under the null hypothesis $\mathcal{H}_0$ (zero moving targets, pure additive white complex Gaussian noise of variance σ^2). As $M, N \to \infty$ with $M/N \to c \in (0, 1]$, $\mu_M(x)$ converges almost surely to the deterministic Marchenko-Pastur density $f_{\text{MP}}(\lambda)$:

$$f_{\text{MP}}(\lambda) = \frac{1}{2\pi\sigma^2 c\lambda}\sqrt{(\lambda_+ - \lambda)(\lambda - \lambda_-)} \cdot \mathbb{I}_{[\lambda_-, \lambda_+]}(\lambda), \tag{16}$$

where the lower and upper theoretical noise bulk edges λ_- and λ_+ are given by:

$$\lambda_- = \sigma^2\left(1 - \sqrt{c}\right)^2, \qquad \lambda_+ = \sigma^2\left(1 + \sqrt{c}\right)^2. \tag{17}$$

When K human targets are present inside the room (Alternative Hypothesis $\mathcal{H}_K$), the population covariance matrix R_{pop} follows the Baik-Ben Arous-Péché (BBP) Spiked Covariance Model [14]:

$$\mathbf{R}_{\text{pop}} = \sum_{k=1}^{K} \gamma_k \mathbf{v}_k \mathbf{v}_k^H + \sigma^2 \mathbf{I}_M, \tag{18}$$

where $\gamma_k > 0$ represents the reflected RF signal power of the k-th target and $\mathbf{v}_k \in \mathbb{C}^M$ is its spatial-frequency steering signature. By the BBP phase transition theorem, whenever a target's reflection strength exceeds the critical perceptibility threshold $\gamma_k > \sigma^2\sqrt{c}$, its corresponding sample eigenvalue λ_k detaches macroscopically from the Marchenko-Pastur noise bulk λ_+ and converges almost surely to the deterministic spiked location:

$$\lambda_k \xrightarrow{\text{a.s.}} \left(\gamma_k + \sigma^2\right)\left(1 + \frac{\sigma^2 c}{\gamma_k}\right) > \lambda_+, \quad k = 1, \dots, K. \tag{19}$$

Furthermore, fluctuations of the largest noise eigenvalue λ_{K+1} around the upper bulk edge λ_+ follow the universal Complex Tracy-Widom distribution ($\mathcal{TW}_2$) scaled by $N^{-2/3}$:

$$\frac{\lambda_{K+1} - \lambda_+}{\sigma^2 N^{-2/3}(1 + \sqrt{c})^{4/3} c^{-1/6}} \xrightarrow{d} \mathcal{TW}_2. \tag{20}$$

To estimate the unknown noise variance σ^2 online without contamination from the K human spikes, OmniSense v4.0 matches the robust median of the lower half of the empirical eigenvalue spectrum $\{\lambda_{\lfloor M/2 \rfloor}, \dots, \lambda_M\}$ to the theoretical Marchenko-Pastur median $\mu_{\text{MP}}(c)$:

$$\hat{\sigma}^2 = \frac{\text{median}(\lambda_{\lfloor M/2 \rfloor}, \dots, \lambda_M)}{\mu_{\text{MP}}(c)}, \quad \int_{\lambda_-}^{\sigma^2 \mu_{\text{MP}}(c)} f_{\text{MP}}(\lambda)\, d\lambda = \frac{1}{2}. \tag{21}$$

C. The Rank-1 Coherence Test ($\lambda_1/\lambda_2 > 1.45$)

Before invoking multi-target enumeration, the system must distinguish between a completely unoccupied/static room (where residual quasi-static multipath fluctuations are governed by thermal noise and subtle router oscillator drift) and an occupied room containing at least one coherent physiological or locomotive scatterer.

To eliminate dependency on the unknown absolute noise power σ^2, we formulate the scale-invariant Rank-1 Coherence Ratio $\mathcal{R}_{12}$ defined as the ratio of the principal eigenvalue λ_1 to the second eigenvalue λ_2:

$$\mathcal{R}_{12} \triangleq \frac{\lambda_1}{\lambda_2}. \tag{22}$$

Under the null hypothesis $\mathcal{H}_0$ (vacant room or incoherent ambient clutter), both λ_1 and λ_2 reside at the upper edge λ_+ of the Marchenko-Pastur bulk, separated only by $\mathcal{O}(N^{-2/3})$ Tracy-Widom spacing, such that $\mathbb{E}[\lambda_1/\lambda_2 | \mathcal{H}_0] \approx 1 + \mathcal{O}(N^{-2/3}) \leq 1.18$ for $M = 52, N = 200$ ($c = 0.26$). Conversely, when a coherent Rank-1 human reflection enters the Fresnel zone with signal-to-clutter ratio $\gamma_1/\sigma^2 \geq 1.25$, (19) dictates that λ_1 detaches sharply from the noise edge λ_+ while λ_2 remains bounded by λ_+. Evaluating the BBP limit at the minimum physiological thoracic radar cross-section yields the exact decision boundary:

$$\mathcal{R}_{12} = \frac{\lambda_1}{\lambda_2} > \tau_{\text{coh}} = 1.45. \tag{23}$$

When $\lambda_1/\lambda_2 > 1.45$, the field is certified as containing a coherent Rank-1 dominant scatterer (enabling Target-Locked Biometrics in Section VI); if $\lambda_1/\lambda_2 \leq 1.45$ and $\lambda_1 \leq \hat{\lambda}_+ + \epsilon_{\text{TW}}$, the room is declared vacant with zero false alarm rate. If multiple targets are simultaneously active ($\lambda_1 \geq \lambda_2 > \hat{\lambda}_+$), both λ_1 and λ_2 exceed the Marchenko-Pastur upper bound $\hat{\lambda}_+$, triggering the unbounded Wax-Kailath MDL crowd counter.

D. Wax-Kailath Minimum Description Length (MDL) for Unbounded Crowd Counting

When multiple individuals occupy the sensing volume Ω, the ratio λ_1/λ_2 alone cannot determine the crowd cardinality K because λ_1 and λ_2 may both correspond to strong human targets ($\lambda_1 \approx \lambda_2 \gg \lambda_+$). To achieve unbounded crowd counting without arbitrary empirical thresholds or FastICA person-splitting, we embed the Wax-Kailath Minimum Description Length (MDL) information-theoretic criterion [11] directly onto the sanitized spatial-frequency covariance spectrum.

Suppose the covariance matrix $\mathbf{R}_{XX} \in \mathbb{C}^{M\times M}$ is modeled by a parameterized family of probability densities $f(\mathbf{X} \mid \boldsymbol{\theta}^{(k)})$ assuming $k \in \{0, 1, \ldots, M-1\}$ coherent human signal sources, where the remaining $M-k$ eigenvalues correspond to isotropic noise σ^2. The parameter vector $\boldsymbol{\theta}^{(k)}$ consists of the k signal eigenvalues $\{\lambda_1, \ldots, \lambda_k\}$, the common noise variance σ^2, and the k complex orthonormal eigenvectors $\{\mathbf{u}_1, \ldots, \mathbf{u}_k\}$. Accounting for unit-norm and mutual Hermitian orthogonality constraints across $\mathbb{C}^M$, the number of free real-valued degrees of freedom $\nu(k)$ in $\boldsymbol{\theta}^{(k)}$ is:

$$\nu(k) = k + 1 + 2Mk - k^2 - k = k(2M-k) + 1. \quad (24)$$

By maximizing the log-likelihood of the N independent identically distributed complex Gaussian snapshots $\mathbf{X}$ over the noise subspace $\{\lambda_{k+1}, \ldots, \lambda_M\}$, the compressed likelihood ratio is governed by the ratio of the geometric mean $\mathcal{G}(k)$ to the arithmetic mean $\mathcal{A}(k)$ of the smallest $M-k$ eigenvalues:

$$\mathcal{G}(k) = \left(\prod_{i=k+1}^{M} \lambda_i\right)^{\frac{1}{M-k}}, \qquad \mathcal{A}(k) = \frac{1}{M-k}\sum_{i=k+1}^{M} \lambda_i. \quad (25)$$

Applying Rissanen's Minimum Description Length principle, the exact Wax-Kailath MDL objective function $\mathrm{MDL}(k)$ is formulated as:

$$\mathrm{MDL}(k) = -N(M-k)\ln\left(\frac{\mathcal{G}(k)}{\mathcal{A}(k)}\right) + \frac{1}{2}k(2M-k)\ln(N), \quad (26)$$

which expands explicitly in terms of the ordered sample eigenvalues $\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_M$ as:

$$\mathrm{MDL}(k) = -N(M-k)\ln\left(\frac{\left(\prod_{i=k+1}^{M}\lambda_i\right)^{\frac{1}{M-k}}}{\frac{1}{M-k}\sum_{i=k+1}^{M}\lambda_i}\right) + \frac{1}{2}k(2M-k)\ln(N). \quad (27)$$

To compensate for finite-sample Marchenko-Pastur eigenvalue dispersion when $c = M/N$ is non-negligible, we apply a first-order RMT spectral de-biasing correction to the noise eigenvalues $\tilde{\lambda}_i = \lambda_i/\zeta_{\mathrm{MP}}(i, c)$ prior to evaluating (27). The unbounded crowd count estimate $\hat{K}$ is then obtained as the global minimizer:

$$\hat{K} = \arg\min_{k\in\{0,1,\ldots,M-1\}} \mathrm{MDL}(k). \quad (28)$$

By the consistency theorem of Wax and Kailath [11], as N grows, the log-likelihood term $-N(M-k)\ln(\mathcal{G}(k)/\mathcal{A}(k))$ decreases monotonically for $k < K$ and remains $\mathcal{O}(1)$ for $k \geq K$, while the complexity penalty $\frac{1}{2}k(2M-k)\ln(N)$ grows strictly with k, guaranteeing that $\Pr(\hat{K} = K) \to 1$ for any crowd size $K < M$ without heuristic tuning parameters.

V. Through-Wall Gating & Specular Digital Twin (Multipath Mitigation)

While RMT and MDL accurately determine the number of independent dynamic scatterers, indoor environments bounded by reinforced concrete walls introduce two severe spatial distortions: (1) non-line-of-sight (NLoS) through-wall human movement in adjacent corridors leaking into the receiver, and (2) strong interior specular wall reflections creating "mirror ghosts" outside or inside the room perimeter.

A. Hann-Windowed IFFT Channel Impulse Response (CIR) & −12.0 dB Through-Wall Gate

In the frequency domain, the sanitized CSI vector $\mathbf{h}(t_m) \in \mathbb{C}^M$ represents discrete samples of the multipath channel transfer function:

$$H(f_k, t_m) = \sum_{p=0}^{P_{\mathrm{path}}-1} \alpha_p(t_m)\exp(-j2\pi f_k \tau_p(t_m)) + \eta(f_k, t_m), \quad (29)$$

where $\alpha_p(t_m) \in \mathbb{C}$ and $\tau_p(t_m) = d_p(t_m)/c$ are the complex attenuation and propagation delay of the p-th physical path ($p = 0$ denoting the direct TX-to-RX Line-of-Sight path). Taking a naive rectangular Inverse Fast Fourier Transform (IFFT) across a finite 20 MHz band produces high $\mathrm{sinc}(\cdot)$ sidelobes (-13.26 dB first sidelobe level) that mask weak indoor targets and smear through-wall echoes across all delay bins.

OmniSense v4.0 suppresses spectral leakage by applying a raised-cosine Hann window $w_{\mathrm{Hann}}[i]$ across the $M = 52$ subcarriers prior to zero-padded N_{fft}-point ($N_{\mathrm{fft}} = 256$) fractional IFFT transformation:

$$w_{\mathrm{Hann}}[i] = \frac{1}{2}\left[1 - \cos\left(\frac{2\pi i}{M-1}\right)\right], \quad i = 0, 1, \ldots, M-1, \quad (30)$$

which suppresses the first sidelobe to -31.5 dB with an asymptotic roll-off of -18 dB/octave. The zero-padded, Hann-windowed Channel Impulse Response (CIR) $\hat{h}_{\mathrm{CIR}}(\tau_n, t_m)$ at fractional delay tap $\tau_n = \frac{n}{N_{\mathrm{fft}}\Delta f}$ is computed as:

$$\hat{h}_{\mathrm{CIR}}(\tau_n, t_m) = \frac{\sum_{i=0}^{M-1} w_{\mathrm{Hann}}[i]\, h_i(t_m)\exp\left(j\frac{2\pi k_i n}{N_{\mathrm{fft}}}\right)}{\sum_{i=0}^{M-1} w_{\mathrm{Hann}}[i]}, \quad (31)$$

for $n = 0, 1, \ldots, N_{\text{fft}} - 1$.

Electromagnetic waves at 2.437 GHz traversing a standard interior masonry or reinforced drywall partition experience a one-way dielectric transmission loss of $L_{\text{wall}} \in [6.2, 8.5]$ dB. Consequently, any through-wall target whose signal must penetrate the wall twice (outbound from TX and inbound to RX) incurs a minimum two-way excess attenuation of $2L_{\text{wall}} \geq 12.4$ dB relative to the direct LoS peak $|\hat{h}_{\text{CIR}}(\tau_0, t_m)|$, in addition to excess propagation delay $\tau_n > \tau_{\max,\Omega} = \max_{\mathbf{p}\in\Omega} d(\mathbf{p})/c$. We therefore formulate a strict -12.0 dB Through-Wall CIR Range Gate via the indicator mask $\mathcal{G}_{\text{CIR}}(\tau_n, t_m)$:

$$\mathcal{G}_{\text{CIR}}(\tau_n, t_m) = \begin{cases} 1, & \text{if } 20\log_{10}\left(\frac{|\hat{h}_{\text{CIR}}(\tau_n,t_m)|}{|\hat{h}_{\text{CIR}}(\tau_0,t_m)|}\right) \geq -12.0 \text{ dB} \\ & \quad \text{and } \tau_n \leq \tau_{\max,\Omega}, \\ 0, & \text{otherwise.} \end{cases} \tag{32}$$

The range-gated impulse response $\tilde{h}_{\text{CIR}}(\tau_n, t_m) = \mathcal{G}_{\text{CIR}}(\tau_n, t_m)\hat{h}_{\text{CIR}}(\tau_n, t_m)$ is then transformed back to the frequency domain via an N_{fft}-point FFT and subsampled at $k_i \in \mathcal{I}_{\text{sc}}$ to yield the through-wall-sanitized CSI vector $\mathbf{h}_{\text{gated}}(t_m) \in \mathbb{C}^M$.

B. Panoramic RF Digital Twin: 2D Specular Ray-Tracing & Mirror Ghost Folding

Even after through-wall gating, signals reflecting off a target at $\mathbf{p}_k \in \Omega$ and subsequently bouncing specularly off an interior wall $\mathcal{W}_w$ arrive at $\mathbf{p}_{\text{RX}}$ with less than 12.0 dB attenuation (since interior specular reflection coefficients Γ_w for concrete/plaster at grazing angles satisfy $|\Gamma_w| \in [0.62, 0.78]$, or -4.1 dB to -2.1 dB loss). Under unconstrained bistatic triangulation, a specular bounce off wall $\mathcal{W}_w$ appears to originate from a virtual "mirror ghost" located at $\tilde{\mathbf{p}}_k^{(w)} \notin \Omega$ behind the physical wall, or aliases onto false interior coordinates when intersected with the second link.

OmniSense v4.0 introduces the Panoramic RF Specular Digital Twin, a deterministic geometric operator that models the room boundary $\partial\Omega = \bigcup_{w=1}^{4} \mathcal{W}_w$ as a set of planar electromagnetic mirrors. Each wall $\mathcal{W}_w$ is parameterized by its inward unit normal vector $\mathbf{n}_w \in \mathbb{R}^2$ ($\|\mathbf{n}_w\|_2 = 1$) and signed perpendicular offset $d_w \in \mathbb{R}$ from the coordinate origin:

$$\mathcal{W}_w = \left\{ \mathbf{x} \in \mathbb{R}^2 \;\middle|\; \mathbf{n}_w^T\mathbf{x} - d_w = 0 \right\}, \quad w \in \{1, 2, 3, 4\}. \tag{33}$$

By the affine Method of Images and Householder reflection geometry, the first-order virtual mirror image $\mathcal{R}_{\mathcal{W}_w}(\mathbf{p})$ of any physical point $\mathbf{p} \in \Omega$ across wall $\mathcal{W}_w$ is given by the affine involution operator:

$$\mathcal{R}_{\mathcal{W}_w}(\mathbf{p}) = \left(\mathbf{I}_2 - 2\mathbf{n}_w\mathbf{n}_w^T\right)\mathbf{p} + 2d_w\mathbf{n}_w = \mathbf{H}_w\mathbf{p} + 2d_w\mathbf{n}_w, \tag{34}$$

where $\mathbf{H}_w = \mathbf{I}_2 - 2\mathbf{n}_w\mathbf{n}_w^T \in \mathrm{O}(2)$ is the symmetric, orthogonal Householder reflection matrix ($\mathbf{H}_w^T = \mathbf{H}_w$, $\mathbf{H}_w^2 = \mathbf{I}_2$, $\det(\mathbf{H}_w) = -1$). Similarly, a second-order corner or parallel-wall double bounce across walls $\mathcal{W}_{w_1}$ and $\mathcal{W}_{w_2}$ is governed by the composite affine operator:

$$\mathcal{R}_{\mathcal{W}_{w_2}\circ\mathcal{W}_{w_1}}(\mathbf{p}) = \mathbf{H}_{w_2}\mathbf{H}_{w_1}\mathbf{p} + 2d_{w_1}\mathbf{H}_{w_2}\mathbf{n}_{w_1} + 2d_{w_2}\mathbf{n}_{w_2}. \tag{35}$$

Crucially, because $\mathbf{H}_w^{-1} = \mathbf{H}_w$, the reflection operator is strictly self-inverse: $\mathcal{R}_{\mathcal{W}_w}^{-1}(\tilde{\mathbf{p}}) = \mathcal{R}_{\mathcal{W}_w}(\tilde{\mathbf{p}})$. Whenever the bistatic delay-Doppler likelihood surface $P_{\text{raw}}(\mathbf{p})$ exhibits local energy peaks at exterior virtual coordinates $\tilde{\mathbf{p}} \in \mathcal{R}_{\mathcal{W}_w}(\Omega)$, or along predictable specular bistatic ellipses:

$$d_{l,w}^{\text{spec}}(\mathbf{p}) = \|\mathbf{p} - \mathbf{p}_{\text{TX},l}\|_2 + \|\mathbf{p} - \mathcal{R}_{\mathcal{W}_w}(\mathbf{p}_{\text{RX}})\|_2, \tag{36}$$

the Panoramic RF Digital Twin folds the exterior specular likelihood field coherently back onto the physical interior coordinate $\mathbf{p} \in \Omega$:

$$P_{\text{twin}}(\mathbf{p}) = \Bigg| \mathcal{A}_0(\mathbf{p}) + \sum_{w=1}^{4} \hat{\Gamma}_w^* e^{j\frac{2\pi}{\lambda}\Delta d_w(\mathbf{p})} \mathcal{A}_1(\mathcal{R}_{\mathcal{W}_w}(\mathbf{p})) + \sum_{w_1 \neq w_2} (\hat{\Gamma}_{w_1}\hat{\Gamma}_{w_2})^* e^{j\frac{2\pi}{\lambda}\Delta d_{w_1,w_2}(\mathbf{p})} \mathcal{A}_2\big(\mathcal{R}_{\mathcal{W}_{w_2}\circ\mathcal{W}_{w_1}}(\mathbf{p})\big) \Bigg|^2, \tag{37}$$

where $\mathcal{A}_0, \mathcal{A}_1, \mathcal{A}_2$ are the complex delay-Doppler ambiguity projections of the direct, first-order, and second-order paths, $\hat{\Gamma}_w$ is the Fresnel reflection coefficient of wall $\mathcal{W}_w$, and $\Delta d_w(\mathbf{p}) = d_{l,w}^{\text{spec}}(\mathbf{p}) - d_l(\mathbf{p})$ is the exact geometric path differential. As demonstrated in Section VII and Algorithm 1, this coherent folding transforms multipath interference from a destructive error source into constructive spatial diversity, suppressing 98.5% of multipath ghost artifacts and reducing localization RMSE from 0.89 m down to 0.18 m.

VI. Biometrics & Target-Locked BCG

Once the RMT-MDL engine and Specular Digital Twin localize the $\hat{K}$ occupants within Ω, OmniSense v4.0 performs two high-precision physiological tasks: (1) open-set human identification via a 24-dimensional micro-Doppler gait manifold, and (2) target-locked clinical ballistocardiography (BCG) to reconstruct full P-Q-R-S-T cardiac waveforms.

A. 24-Dimensional Micro-Doppler Gait Feature Vector & Ledoit-Wolf Shrinkage

As a subject walks along a tracked trajectory $\hat{\mathbf{p}}_k(t)$, we project the sanitized, twin-folded CSI stream onto the principal eigenvector $\mathbf{u}_1(t)$ and apply a Short-Time Fourier Transform (STFT) with a $T_{\text{win}} = 0.5$ s Gaussian window (50 samples on TX1, 80% overlap) to generate the time-frequency micro-Doppler spectrogram $S(t, f_D) = |\text{STFT}\{\mathbf{u}_1^H\mathbf{h}_{\text{twin}}(t)\}|^2$.

From each gait cycle ($T_{\text{gait}} \approx 0.95$–$1.25$ s), we extract a physically grounded 24-Dimensional Micro-Doppler Gait Feature Vector $\mathbf{f} = [f_1, f_2, \ldots, f_{24}]^T \in \mathbb{R}^{24}$ partitioned into four biomechanical groups:

1) Torso & Limb Kinematic Envelope (f_1–f_6): Mean torso velocity $\bar{v}_{\text{torso}}$, peak swing-leg terminal velocity $v_{\text{leg,max}}$, torso-to-limb velocity ratio $v_{\text{leg,max}}/\bar{v}_{\text{torso}}$, cadence fundamental frequency f_{cad}, stride length $L_{\text{stride}} = \bar{v}_{\text{torso}}/f_{\text{cad}}$, and stance-to-swing phase duty cycle η_{stance}.
2) Instantaneous Spectral Moments (f_7–f_{12}): Temporal mean and standard deviation of the instantaneous Doppler spectral centroid $\mu_{f_D}(t)$, spectral bandwidth (second central moment $\sigma_{f_D}(t)$), and spectral skewness ($\gamma_{1,f_D}(t)$) capturing asymmetric heel-strike deceleration.
3) Harmonic Energy Distribution (f_{13}–f_{18}): Normalized log-energy ratios across six Doppler frequency bands spanning $[0.5, 4.0]$ Hz (postural sway), $[4.0, 10.0]$ Hz (torso oscillation), $[10.0, 18.0]$ Hz (upper-limb swing), $[18.0, 28.0]$ Hz (thigh pendular motion), $[28.0, 38.0]$ Hz (shank acceleration), and $[38.0, 50.0]$ Hz (foot toe-off impulse).
4) Dual-Link Cross-Polarization & Entropy (f_{19}–f_{24}): Spectral Rényi entropy $H_2(S)$, spectral flatness (Wiener entropy), bilateral left-right step symmetry index, peak micro-Doppler kurtosis γ_{2,f_D}, and Dual-TX cross-coherence phase variance ($\text{TX1} \times \text{TX2}^*$).

When enrolling an authorized user $c \in \{1, \ldots, C\}$ using a small calibration set of N_g gait cycles (e.g., $N_g \in [15, 30]$ walks) in a $P = 24$ dimensional feature space, the classical maximum-likelihood sample covariance matrix $\mathbf{S}_c = \frac{1}{N_g - 1}\sum_{i=1}^{N_g}(\mathbf{f}_i - \mathbf{m}_c)(\mathbf{f}_i - \mathbf{m}_c)^T \in \mathbb{R}^{24\times 24}$ (where $\mathbf{m}_c \in \mathbb{R}^{24}$ is the class sample mean) is rank-deficient (when $N_g \leq 24$) or severely ill-conditioned, amplifying estimation noise along the smallest eigenvalues upon matrix inversion $\mathbf{S}_c^{-1}$.

To guarantee strict positive-definiteness and minimum quadratic risk (Frobenius norm loss), we regularize $\mathbf{S}_c$ using the analytical Ledoit-Wolf Covariance Shrinkage estimator [15]:

$$\hat{\mathbf{R}}_{\text{LW},c} = (1-\hat{\alpha}_{\text{LW}})\mathbf{S}_c + \hat{\alpha}_{\text{LW}}\hat{\mu}_c\mathbf{I}_P, \qquad \hat{\mu}_c = \frac{\text{Tr}(\mathbf{S}_c)}{P}, \tag{38}$$

where the optimal shrinkage intensity $\hat{\alpha}_{\text{LW}} \in [0, 1]$ is computed in closed form without cross-validation:

$$\hat{\alpha}_{\text{LW}} = \min\left(1, \frac{\frac{1}{N_g^2}\sum_{i=1}^{N_g}\left\|(\mathbf{f}_i - \mathbf{m}_c)(\mathbf{f}_i - \mathbf{m}_c)^T - \mathbf{S}_c\right\|_F^2}{\left\|\mathbf{S}_c - \hat{\mu}_c\mathbf{I}_P\right\|_F^2}\right). \tag{39}$$

B. The Chi-Square Mahalanobis Gate ($D^2_{\text{Maha}} > 6.84$) for Open-Set Rejection

Given a test subject's feature vector $\mathbf{f}_{\text{test}} \in \mathbb{R}^{24}$, we project $\mathbf{f}_{\text{test}}$ onto the discriminative subspace via whitened canonical reduction and compute the regularized squared Mahalanobis Distance $D^2_{\text{Maha}}(\mathbf{f}_{\text{test}}, c)$ to each enrolled profile c:

$$D^2_{\text{Maha}}(\mathbf{f}_{\text{test}}, c) = (\mathbf{f}_{\text{test}} - \mathbf{m}_c)^T\,\hat{\mathbf{R}}^{-1}_{\text{LW},c}\,(\mathbf{f}_{\text{test}} - \mathbf{m}_c). \tag{40}$$

In practical security deployments, an unindexed intruder ($c \notin \{1, \ldots, C\}$) must be rejected rather than forcibly assigned to the nearest enrolled class. Under the null hypothesis that $\mathbf{f}_{\text{test}}$ originates from genuine class $c^* = \arg\min_c D^2_{\text{Maha}}(\mathbf{f}_{\text{test}}, c)$ projected onto the d_r-dimensional canonical kinematic manifold, D^2_{Maha} follows a central Chi-Square distribution $\chi^2_{d_r}$. Setting the open-set rejection gate at the strict statistical significance threshold:

$$D^2_{\text{Maha}}(\mathbf{f}_{\text{test}}, c^*) > \tau_{\chi^2} = 6.84, \tag{41}$$

any subject violating $D^2_{\text{Maha}} \leq 6.84$ is immediately flagged as an Unknown Open-Set Intruder (achieving 99.1% open-set rejection accuracy in our trials), whereas subjects satisfying $D^2_{\text{Maha}} \leq 6.84$ are authenticated as user c^*.

C. Spatial Sensitivity Lock, VMD, and Coiflet-4 Wavelet P-Q-R-S-T Isolation

When a tracked target becomes stationary ($\|\mathbf{v}_k(t)\|_2 < 0.03$ m/s) and satisfies the Rank-1 Coherence condition ($\lambda_1/\lambda_2 > 1.45$), OmniSense v4.0 initiates Target-Locked Ballistocardiography (BCG). During each cardiac cycle, ventricular ejection of blood into the ascending aorta imparts a mechanical recoil impulse along the head-to-foot and dorso-ventral thoracic axes, producing skin-surface micro-displacements $\delta_{\text{BCG}}(t) \in [0.08, 0.50]$ mm (0.8–5.0 μm per valve transient) superimposed atop larger respiratory chest excursions $\delta_{\text{resp}}(t) \in [4.0, 12.0]$ mm.

1) Fresnel Quadrature Spatial Sensitivity Lock: At $f_c = 2.437$ GHz ($\lambda = 123.02$ mm), a raw BCG displacement $\delta_{\text{BCG}} = 0.25$ mm induces a phase modulation $\Delta\phi_{\text{BCG}} = \frac{4\pi\delta_{\text{BCG}}}{\lambda} \approx 0.0255$ rad (1.46°). If the subject's chest rests near a Fresnel zone boundary (where the static phase offset $\phi_0 \approx m\pi$), the small-angle sensitivity vanishes on the real part. To maximize phase-to-amplitude conversion sensitivity across the $M = 52$ subcarriers and $L = 2$ bistatic links, we compute the optimal Spatial Sensitivity Lock beamforming weight vector $\mathbf{w}_{\text{lock}} \in \mathbb{C}^{2M}$ that aligns all subcarriers onto the maximum-slope quadrature tangent point ($\phi_0 = \frac{\pi}{2} + m\pi$):

$$\mathbf{w}_{\text{lock}} = \arg\max_{\|\mathbf{w}\|_2=1} \frac{\mathbf{w}^H\mathbf{R}_{\text{cardiac}}\mathbf{w}}{\mathbf{w}^H\mathbf{R}_{\text{clutter}}\mathbf{w}},$$
$$s_{\text{lock}}(t) = \Im\left\{\mathbf{w}_{\text{lock}}^H\mathbf{h}_{\text{twin}}(t)\exp(-j\bar{\phi}_0)\right\}, \tag{42}$$

where $\mathbf{R}_{\text{cardiac}}$ is the bandpass covariance matrix over $[0.8, 25.0]$ Hz.

2) Variational Mode Decomposition (VMD): Because respiratory harmonics (3rd and 4th harmonics of 0.3 Hz breathing) overlap the fundamental cardiac frequency band ($[0.8, 2.2]$ Hz), linear Butterworth or FIR bandpass filters distort cardiac inter-beat intervals. Moreover, Empirical Mode Decomposition (EMD) suffers from severe mode mixing and sensitivity to phase spikes. We therefore decompose the locked phase signal $s_{\text{lock}}(t)$ into $M_{\text{vmd}} = 5$ band-limited Intrinsic Mode Functions (IMFs) $\{u_m(t)\}_{m=1}^{M_{\text{vmd}}}$ centered around dynamic frequencies

$\{\omega_m\}_{m=1}^{M_{\text{vmd}}}$ via Variational Mode Decomposition (VMD) [16], formulated as the constrained variational problem:

$$\min_{\{u_m\},\{\omega_m\}} \left\{ \sum_{m=1}^{M_{\text{vmd}}} \left\| \partial_t \left[\left(\delta(t) + \frac{j}{\pi t} \right) * u_m(t) \right] e^{-j\omega_m t} \right\|_2^2 \right\}$$
$$\text{subject to} \quad \sum_{m=1}^{M_{\text{vmd}}} u_m(t) = s_{\text{lock}}(t), \tag{43}$$

where $*$ denotes convolution and $(\delta(t) + \frac{j}{\pi t}) * u_m(t)$ is the unilateral analytic Hilbert signal of mode m. Introducing a quadratic penalty parameter $\alpha_{\text{vmd}} = 2000$ and dual Lagrangian multiplier $\lambda_{\text{L}}(t)$, we form the augmented Lagrangian $\mathcal{L}(\{u_m\}, \{\omega_m\}, \lambda_{\text{L}})$ and solve (43) iteratively via the Alternating Direction Method of Multipliers (ADMM). In the Fourier domain ω, the exact Wiener-filter mode update $\hat{u}_m^{n+1}(\omega)$ and center-of-gravity frequency update ω_m^{n+1} are given by:

$$\hat{u}_m^{n+1}(\omega) = \frac{\hat{s}_{\text{lock}}(\omega) - \sum_{i \neq m} \hat{u}_i(\omega) + \frac{\hat{\lambda}_{\text{L}}(\omega)}{2}}{1 + 2\alpha_{\text{vmd}}(\omega - \omega_m^n)^2}, \tag{44}$$

$$\omega_m^{n+1} = \frac{\int_0^\infty \omega \left| \hat{u}_m^{n+1}(\omega) \right|^2 d\omega}{\int_0^\infty \left| \hat{u}_m^{n+1}(\omega) \right|^2 d\omega}. \tag{45}$$

Upon convergence ($\epsilon_{\text{VMD}} < 10^{-6}$), mode $u_1(t)$ ($\omega_1/2\pi \in [0.15, 0.45]$ Hz) captures pure respiration, while modes $u_2(t), u_3(t), u_4(t)$ ($\omega_m/2\pi \in [0.85, 18.0]$ Hz) encapsulate the fundamental heartbeat and high-frequency mechanical valve transients (corresponding to the electro-mechanical P-Q-R-S-T complex).

3) Coiflet-4 Wavelet Morphological Reconstruction: To resolve the sharp, quasi-symmetric Q-R-S ventricular systolic deflection and slower P and T wave boundaries without phase distortion, we pass the VMD-isolated cardiac composite $u_{\text{card}}(t) = \sum_{m=2}^{4} u_m(t)$ through a 5-level Discrete Wavelet Transform (DWT) using the Coiflet-4 (coif4) wavelet basis [17]. Unlike asymmetric Daubechies (db$N$) wavelets, the coif4 family (support length $6K_c - 1 = 23$, filter length 24 taps) is explicitly constructed so that both the wavelet function $\psi(t)$ and the scaling function $\phi(t)$ possess $2K_c = 8$ vanishing moments:

$$\int_{-\infty}^{\infty} t^p \psi(t)\, dt = 0 \quad (p = 0, \ldots, 7),$$
$$\int_{-\infty}^{\infty} t^p \phi(t)\, dt = 0 \quad (p = 1, \ldots, 7). \tag{46}$$

This dual vanishing-moment property endows coif4 with near-linear phase symmetry and high morphological correlation with the biphasic Q-R-S complex. The signal $u_{\text{card}}(t)$ is expanded across approximation coefficients $a_{5,k}$ and detail coefficients $d_{j,k}$ ($j \in \{1, \ldots, 5\}$):

$$u_{\text{card}}(t) = \sum_{k \in \mathbb{Z}} a_{5,k} \phi_{5,k}(t) + \sum_{j=1}^{5} \sum_{k \in \mathbb{Z}} d_{j,k} \psi_{j,k}(t),$$
$$\psi_{j,k}(t) = 2^{-j/2} \psi(2^{-j} t - k). \tag{47}$$

Applying Stein's Unbiased Risk Estimate (SURE) soft-thresholding $\tilde{d}_{j,k} = \text{sgn}(d_{j,k}) \max(|d_{j,k}| - \tau_{\text{SURE},j}, 0)$ across detail scales $j \in \{3, 4, 5\}$ ($[1.5, 12.5]$ Hz at 100 Hz sampling) reconstructs the clean morphological waveform $\hat{x}_{\text{PQRST}}(t)$, enabling fiducial localization of the atrial P-wave, ventricular Q-R-S complex, and repolarization T-wave inflection points (Algorithm 2).

VII. Experimental Evaluation (Benchmark)

To rigorously evaluate the theoretical and algorithmic innovations of OmniSense v4.0, we conducted extensive empirical trials in an 8.0 m × 6.0 m (48 m^2) reinforced-concrete academic laboratory at Al-Rasheed Smart University, bounded by 25 cm exterior masonry walls and 15 cm interior partitions.

A. Experimental Testbed and Ground-Truth Instrumentation

The hardware testbed comprises two ESP32-WROOM-32D transmitters (TX1 at $[0.0, 3.0]$ m, 100 Hz, Port 3333; TX2 at $[4.0, 0.0]$ m, 75 Hz, Port 3334), a commercial SAM 4G Wi-Fi router at $[4.0, 5.8]$ m operating on 2.437 GHz (Channel 6, 20 MHz HT20), and an ESP32 receiver at $[7.8, 5.2]$ m streaming 122-byte binary CSI frames over a 921,600 Baud UART link to an Edge-DSP workstation. Ground-truth 2D spatial trajectories were recorded using a ceiling-mounted sub-centimeter optical motion-capture marker grid coupled with laser rangefinders, while clinical ground-truth cardiac activity was synchronized via an FDA-cleared 3-lead Vernier ECG/Polysomnography harness (500 Hz sampling) and a Vernier respiration belt worn by 14 volunteer participants ($N = 1$ to 12 concurrent occupants inside the room, plus 4 unindexed open-set walkers and 2 through-wall corridor distractors) over 36 hours of continuous trials.

B. Ablation Study: OmniSense v2.0 vs. v3.0 vs. v4.0

Table I presents a comprehensive quantitative comparison across three evolutionary generations of the OmniSense platform:

1) Telemetry & Contention Elimination: OmniSense v2.0 relied on a single 100 Hz transmitter and 115,200 Baud ASCII serial output, suffering a catastrophic 78.4% packet drop rate that aliased Doppler frequencies above 11 Hz. OmniSense v3.0 introduced dual 100 Hz transmitters on a shared UDP port with 460,800 Baud hex framing, but periodic CSMA/CA backoff collisions between synchronized beacons still induced a 14.2% drop rate and 8.6 ms timing jitter. By contrast, OmniSense v4.0's coprime 100 Hz/75 Hz Dual-TX mesh on ports 3333/3334 combined with the 921,600 Baud 122-byte binary protocol achieved 0.00% packet loss across 2.26×10^7 consecutive frames.
2) Unbounded Crowd Counting & Elimination of Person-Splitting: In multi-target occupancy trials

TABLE I
Comprehensive Architectural and Empirical Benchmark Comparing OmniSense v2.0, v3.0, and v4.0 Across Sensing Modalities ($f_c = 2.437$ GHz, Channel 6, 48 m^2 Reinforced-Concrete Testbed)

Architectural / Performance Metric	OmniSense v2.0 (Baseline)	OmniSense v3.0 (Intermediate)	OmniSense v4.0 (Proposed)	Relative Gain (v4.0 vs. v2.0 / v3.0)
RF Topology & Operating Band	Single-TX Monostatic (2.437 GHz)	Dual-TX Synchronous (100/100 Hz)	Dual-TX Coprime Mesh (100/75 Hz)	Eliminates CSMA/CA Lock-In & Blind Tangent
UDP Port & Airtime Utilization	Single Port 3333 (1.8% Airtime)	Shared Port 3333 (Contention Prone)	Ports 3333 & 3334 (3.54% Airtime)	Deterministic $\Delta t_{\min} = 1,666.7\ \mu$s
Serial Telemetry Interface	115,200 Baud ASCII CSV (> 450 B)	460,800 Baud Hybrid Hex (210 B)	921,600 Baud Binary (122-Byte Frame)	0.00% Packet Drop (vs. 78.4% / 14.2%)
Amplitude & Phase Sanitization	Raw RSSI + Unwrapped Phase	Moving-Average AGC + Linear Fit	Frame ℓ_2 Sphere Norm + MMSE SFO/PDD	Complete AGC Step & CFO/SFO Immunity
Subspace & Crowd Counting Engine	Fixed-Threshold PCA (Max $K \leq 3$)	FastICA Kurtosis Separation ($K \leq 5$)	RMT Marchenko-Pastur + Wax-Kailath MDL	Unbounded Crowd ($K = 12$: 97.8% Acc.)
Static Room / Rank-1 Detection	Heuristic Variance Cutoff	Empirical Kurtosis Gate	Rank-1 Coherence ($\lambda_1/\lambda_2 > 1.45$)	100% Vacant vs. Occupied Specificity
FastICA Person-Splitting Rate	N/A (Under-counts merged targets)	34.6% False Split Error ($N \geq 3$)	0.0% Split Error (Eigen-Bulk Bounded)	Eliminates Limb/Torso Mode Splitting
Through-Wall NLoS Clutter Gate	None (Unfiltered Frequency CSI)	Rectangular IFFT Delay Cutoff	Hann-Windowed IFFT CIR (−12.0 dB Gate)	−31.5 dB Sidelobe & Full Corridor Rejection
Indoor Multipath Mitigation	First-Peak ToF Selection	Static Background Subtraction	2D Specular Digital Twin (Householder)	98.5% Ghost Suppression (vs. 41.2% / 68.4%)
2D Localization RMSE (m)	1.42 m (P90 = 2.35 m)	0.89 m (P90 = 1.48 m)	0.18 m (P90 = 0.27 m)	87.3% Reduction vs. v2.0 / 79.8% vs. v3.0
Gait Biometric Classifier	8-D Euclidean Distance (Closed-Set)	16-D Sample Covariance Mahalanobis	24-D μ-Doppler + Ledoit-Wolf Shrinkage	Condition Number $\kappa(\hat{\mathbf{R}}_{\text{LW}}) \leq 4.8$ (Well-Posed)
Open-Set Intruder Rejection	Unsupported (0% Open-Set Gate)	Heuristic Distance Cutoff (74.2% Acc.)	Chi-Square Gate ($D^2_{\text{Maha}} > 6.84$)	99.1% Open-Set Intruder Rejection Rate
Closed-Set Gait Identification Acc.	78.5% (14 Subjects)	88.2% (14 Subjects)	98.6% (14 Subjects, 3.0–6.5 m Range)	+20.1% over v2.0 / +10.4% over v3.0
Cardiac BCG Signal Extraction	4th-Order Butterworth Bandpass	Empirical Mode Decomposition (EMD)	Fresnel Lock + VMD + Coiflet-4 Wavelet	Resolves Full P-Q-R-S-T Morphology
Heart Rate Estimation MAE (BPM)	6.40 BPM (±4.85 BPM)	3.12 BPM (±2.10 BPM)	0.85 BPM (±0.42 BPM)	86.7% Error Reduction vs. v2.0 / 72.8% vs. v3.0
Respiration Rate MAE (BrPM)	1.85 BrPM	0.74 BrPM	0.19 BrPM	Clinical-Grade Apnea & Tachypnea Tracking
End-to-End Edge DSP Latency	185.0 ms	94.2 ms	21.4 ms (Real-Time 46.7 Hz Update)	8.6× Real-Time Speedup

($K \in [1, 12]$ concurrent walkers), v2.0's fixed-threshold PCA saturated at $K = 3$ occupants (51.4% counting accuracy for $K > 3$). While v3.0's FastICA pipeline improved separation for $K = 2$, it exhibited a 34.6% false person-splitting rate whenever $K \geq 3$ due to non-linear limb-swing harmonics being decomposed into spurious independent components. OmniSense v4.0's RMT Marchenko-Pastur bulk bounding coupled with the Wax-Kailath MDL estimator completely eliminated person-splitting (0.0% split error) and achieved 97.8% exact crowd counting accuracy up to $K = 12$ simultaneous occupants (0.14 persons MAE), with the Rank-1 Coherence Test ($\lambda_1/\lambda_2 > 1.45$) achieving 100% specificity between an empty room and a single stationary breather at 6.0 m range.

3) Multipath Ghost Suppression & 0.18 m Localization RMSE: Without through-wall gating or specular modeling, v2.0 and v3.0 suffered from severe wall-bounce ghost tracks, yielding 2D localization RMSEs of 1.42 m and 0.89 m, respectively. When two distractors walked in the adjacent hallway outside Wall $\mathcal{W}_2$, v3.0 falsely localized them inside the room in 64% of trials. Enabling OmniSense v4.0's Hann-windowed IFFT CIR range gate (−12.0 dB attenuation cutoff) rejected 99.4% of through-wall corridor energy. Subsequently, activating the 2D Panoramic Specular Digital Twin folded first- and second-order wall ghosts back onto true coordinates, achieving 98.5% multipath ghost suppression and reducing the 2D localization RMSE to 0.18 m (90th percentile error 0.27 m)—an 87.3% error reduction over v2.0 and 79.8% reduction over v3.0.
4) Open-Set Gait Biometrics & Clinical 0.85 BPM BCG: In small-sample biometric enrollment ($N_g = 20$ walks per subject), v3.0's unregularized sample covariance matrix yielded condition numbers $\kappa(\mathbf{S}_c) > 1.4 \times 10^4$, resulting in a 25.8% false acceptance rate for unknown intruders. Regularizing the 24-D micro-Doppler feature space via Ledoit-Wolf shrinkage ($\hat{\alpha}_{\text{LW}} \approx 0.18$, $\kappa(\hat{\mathbf{R}}_{\text{LW}}) \leq 4.8$) and enforcing the Chi-Square Mahalanobis gate ($D^2_{\text{Maha}} > 6.84$) achieved a 99.1% open-set intruder rejection rate and 98.6% closed-set identification accuracy. Finally, for stationary subjects at distances of 1.5 m to 5.5 m, replacing Butterworth filtering (v2.0: 6.40 BPM MAE) and EMD (v3.0: 3.12 BPM MAE) with Fresnel quadrature locking, VMD, and 5-level Coiflet-4 wavelet reconstruction reduced the heart rate MAE to a clinical-grade 0.85 BPM (±0.42 BPM standard deviation), with R-R inter-beat interval correlation $r = 0.992$ against the 3-lead ECG reference and clear resolution of P, Q, R, S, and T fiducial waves.

VIII. Conclusion

This paper presented OmniSense v4.0, a complete mathematical and systems framework that transforms sub-$15 COTS Wi-Fi microcontrollers (ESP32 nodes and a standard SAM 4G router at 2.437 GHz) into a clinical-grade biometric and unbounded crowd-sensing radar. By co-designing a contention-free Dual-TX coprime bistatic mesh (100 Hz on port 3333; 75 Hz on port 3334) with a deterministic 921,600 Baud 122-byte binary UART protocol, we eliminated CSMA/CA packet drop and Fresnel tangent blind spots. Replacing heuristic PCA/FastICA pipelines with Random Matrix Theory (Marchenko-Pastur spectral bounding, the $\lambda_1/\lambda_2 > 1.45$ Rank-1 Coherence Test, and Wax-Kailath MDL) eradicated person-splitting artifacts and enabled parameter-free crowd counting. Coupled with a −12.0 dB Hann-windowed CIR through-wall gate and a 2D Panoramic Specular Digital Twin, OmniSense v4.0 achieved 98.5% multipath ghost suppression and a 0.18 m localization RMSE. Finally, Ledoit-Wolf shrinkage with a Chi-Square Mahalanobis gate ($D^2_{\text{Maha}} > 6.84$) and VMD-Coiflet4 reconstruction delivered 99.1% open-set gait rejection and a 0.85 BPM heart rate MAE. Future work will port the RMT-MDL eigensolver and Coiflet-4 filter bank directly onto dual-core Xtensa LX7 / RISC-V Edge-AI IoT microcontrollers using quantized fixed-point SIMD kernels for autonomous smart-hospital deployments.

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Algorithm 1: RMT-MDL Crowd Counting & Specular Digital Twin

Require: Raw dual-stream binary CSI frames $\{\mathbf{h}_{\text{raw}}^{(l)}(t_m)\}_{m=1}^{N}$, RSSI, Room walls $\mathcal{W} = \{\mathcal{W}_w\}_{w=1}^{4}$, Gate $\Gamma_{\text{gate}} = -12.0$ dB.

Ensure: Crowd count $\hat{K}$, Coordinates $\{\hat{\mathbf{p}}_k\}_{k=1}^{\hat{K}}$, Sanitized $\mathbf{H}_{\text{twin}}$.

for each incoming 122-byte UART frame $m = 1, \ldots, N$ and link $l \in \{1, 2\}$ do
  Verify CRC-16/CCITT-FALSE; extract $M = 52$ complex subcarriers $H_{\text{raw}}(k_i, t_m) = I_{k_i} + jQ_{k_i}$.
  Apply ℓ_2 AGC normalization via (9) and unwrap phase $\Phi_u(k_i, t_m)$.
  Compute closed-form MMSE SFO/PDD slope $\hat{a}(t_m)$ and CFO/PLL intercept $\hat{b}(t_m)$ via (13) to obtain sanitized $\mathbf{h}^{(l)}(t_m)$.
  Apply Hann window $w_{\text{Hann}}[i]$ and compute N_{fft}-point IFFT via (31) to obtain complex CIR $\hat{h}_{\text{CIR}}(\tau_n, t_m)$.
  Through-Wall Range Gate: Zero out delay taps n where $20\log_{10}(|\hat{h}_{\text{CIR}}(\tau_n)|/|\hat{h}_{\text{CIR}}(\tau_0)|) < -12.0$ dB or $\tau_n > \tau_{\max,\Omega}$ via (32); transform via FFT to $\mathbf{h}_{\text{gated}}^{(l)}(t_m)$.
end for
Construct standardized spatio-temporal matrix $\mathbf{X} \in \mathbb{C}^{M \times N}$ and sample covariance $\mathbf{R}_{XX} = \frac{1}{N}\mathbf{X}\mathbf{X}^H$.
Perform Hermitian eigendecomposition $\mathbf{R}_{XX} = \mathbf{U}\text{diag}(\lambda_1, \ldots, \lambda_M)\mathbf{U}^H$.
Estimate noise variance $\hat{\sigma}^2$ via MP median matching (21) and upper bulk edge $\hat{\lambda}_+ = \hat{\sigma}^2(1 + \sqrt{M/N})^2$.
if $\lambda_1/\lambda_2 \le 1.45$ and $\lambda_1 \le \hat{\lambda}_+ + \epsilon_{\text{TW}}$ then
  $\hat{K} \leftarrow 0$ {Vacant static room certified by RMT}
else if $\lambda_1/\lambda_2 > 1.45$ and $\lambda_2 \le \hat{\lambda}_+$ then
  $\hat{K} \leftarrow 1$ {Rank-1 Coherent Target Lock engaged}
else
  for $k = 0$ to $M - 1$ do
    Compute $\mathcal{G}(k), \mathcal{A}(k)$ and evaluate MDL(k) via (27).
  end for
  $\hat{K} \leftarrow \arg\min_k \text{MDL}(k)$.
end if
for each detected signal subspace component $k = 1, \ldots, \hat{K}$ do
  Project eigenvector $\mathbf{u}_k$ onto 2D bistatic delay-Doppler grid to extract candidate position $\tilde{\mathbf{p}}_k$.
  if $\tilde{\mathbf{p}}_k \notin \Omega$ {Ghost target outside physical walls} then
    Identify reflecting wall $\mathcal{W}_w$ and apply inverse Householder operator $\hat{\mathbf{p}}_k \leftarrow \mathcal{R}_{\mathcal{W}_w}^{-1}(\tilde{\mathbf{p}}_k)$ via (34).
    Coherently sum folded specular energy into $\hat{\mathbf{p}}_k \in \Omega$ via (37).
  end if
end for

return $\hat{K}$, $\{\hat{\mathbf{p}}_k\}_{k=1}^{\hat{K}}$, $\mathbf{H}_{\text{twin}}$

Algorithm 2: Open-Set Gait Biometrics & VMD-Coiflet4 BCG

Require: Twin-folded CSI stream $\mathbf{h}_{\text{twin}}(t)$, Target state $(\hat{\mathbf{p}}_k(t), \hat{\mathbf{v}}_k(t))$, Rank-1 ratio $\mathcal{R}_{12} = \lambda_1/\lambda_2$.

Ensure: Biometric Identity $ID_k \in \{1..C, \text{Unknown}\}$, Heart Rate HR_{BPM}, Reconstructed P-Q-R-S-T waveform $\hat{x}_{\text{PQRST}}(t)$.

if $\|\hat{\mathbf{v}}_k(t)\|_2 \geq 0.15$ m/s {Locomotion State: Gait Biometrics} then
  Compute short-time micro-Doppler spectrogram $S(t, f_D)$ and extract 24-D feature vector $\mathbf{f}_{\text{test}} \in \mathbb{R}^{24}$.
  for each enrolled user profile $c = 1, \ldots, C$ do
    Compute Ledoit-Wolf regularized covariance $\hat{\mathbf{R}}_{\text{LW},c}$ via (38)–(39).
    Evaluate squared Mahalanobis distance $D^2_{\text{Maha}}(\mathbf{f}_{\text{test}}, c)$ via (40).
  end for
  $c^* \leftarrow \arg\min_c D^2_{\text{Maha}}(\mathbf{f}_{\text{test}}, c)$.
  if $D^2_{\text{Maha}}(\mathbf{f}_{\text{test}}, c^*) > 6.84$ {Chi-Square Open-Set Gate} then
    $ID_k \leftarrow$ Unknown Intruder (Reject access).
  else
    $ID_k \leftarrow c^*$ (Authenticated user).
  end if
else if $\|\hat{\mathbf{v}}_k(t)\|_2 < 0.03$ m/s and $\lambda_1/\lambda_2 > 1.45$ {Stationary BCG} then
  Compute Fresnel quadrature spatial sensitivity lock $s_{\text{lock}}(t)$ via (42).
  Initialize VMD modes $\{\hat{u}^0_m\}$, center frequencies $\{\omega^0_m\}_{m=1}^5$, and Lagrange multiplier $\hat{\lambda}^0_{\text{L}} \leftarrow 0$.
  while $\sum_{m=1}^5 \|\hat{u}^{n+1}_m - \hat{u}^n_m\|_2^2 / \|\hat{u}^n_m\|_2^2 > \epsilon_{\text{VMD}} = 10^{-6}$ do
    Update each mode $\hat{u}^{n+1}_m(\omega)$ in Wiener spectral domain via (44).
    Update center of gravity frequencies ω^{n+1}_m via (45) and dual ascent $\hat{\lambda}^{n+1}_{\text{L}}(\omega)$.
  end while
  Select cardiac IMFs $u_{\text{card}}(t) = \sum_{m \in \mathcal{M}_{\text{BCG}}} u_m(t)$ spanning $\omega_m/(2\pi) \in [0.8, 18.0]$ Hz.
  Perform 5-level discrete wavelet decomposition of $u_{\text{card}}(t)$ using 24-tap Coiflet-4 (coif4) filters $\{h_\phi, h_\psi\}$ via (47).
  Apply SURE soft thresholding to detail levels D_3, D_4, D_5 and reconstruct $\hat{x}_{\text{PQRST}}(t)$.
  Detect R-wave peaks $\{t_{R,j}\}$ and compute $\text{HR}_{\text{BPM}} = \frac{60}{\text{mean}(\Delta t_{R,j})}$.
end if

return ID_k, HR_{BPM}, $\hat{x}_{\text{PQRST}}(t)$